Personal Learning Journey – Thinila N.A.D.I.D.

Student	Student ID	Group	Member	Role
Thinila N.A.D.I.D.	IT24102032	2026-AI-33	Member 1	Problem Framing & Data Understanding

My assigned role is to establish what the student data represents and identify evidence that should guide preprocessing and modelling. The central lesson from this stage is that a useful pattern must be interpreted in its data and prediction-time context.

Contribution and evidence

The notebook associated with my role describes the Mathematics dataset, its source, student-level rows and 33 variables. It checks distributions, categories, missing values, duplicates and possible outliers, then examines relationships with final grade G3. The recorded scope is 395 students, with no missing cells or exact duplicate full rows. G3 is an integer-valued outcome on a 0–20 scale, although the project predicts a continuous grade.

A decision that shaped the project

The 38 zero final grades required more careful reasoning than simply treating them as unusual observations. Their causes are unknown, so the current project retains them. Excluding them would remove students relevant to the support objective and change the population being evaluated. Strong associations between G1/G2 and G3 also motivated separate feature scenarios; correlation alone does not establish leakage, because availability at the prediction date is what matters.

Learning and critical judgement

This work shows why statistical summaries, plots and variable definitions must be considered together. Absences have a long upper tail, but a large value is not automatically an error. Group differences and correlations suggest questions rather than prove causes. A remaining notebook issue is that the demographic plotting loop repeats school on the horizontal axis; those plots need correction before being used as evidence for other variables.

Team handoff and next steps

My role supplies the next stage with variable meanings, data-quality findings and questions about zeros, prior grades and absences. Member 2 can use these findings to justify transformations rather than apply a cleaning recipe. A limitation is that EDA examined the full data before splitting, so the later holdout is not fully blind to exploration. My next priorities would be accurate plot labels and clearer separation of observed evidence from hypotheses.

Personal reflection

At first I read the zero final grades as probable recording errors. Checking them against other columns changed that view: all 38 students have a recorded G1 between 4 and 12, and all 38 have zero recorded absences. The pattern suggests a consistent recording practice rather than blank rows, but it does not reveal the cause, so I argued for keeping them and stating the uncertainty. I framed the problem around staff-reviewed support and checked the dictionary codes against data/student.txt, where the failures-code conflict appeared.

Evidence and AI use

Evidence: notebooks/01_data_understanding_eda.ipynb. I used an AI assistant to explain EDA concepts, such as skewness, correlation strength and why the zero grades make the G3 distribution bimodal. I checked each explanation against the notebook's own statistics and plots before using it in my interpretation. AI also helped draft this report; its figures were checked against the notebook outputs.